

Inflationary Observables from Dimensional Flow in Scale-Dependent Geometric Field Theory

David A. Besemer

Independent Researcher

david.besemer@sdgft.org

arXiv:SDGFT-2026-17

Abstract. We derive the inflationary observables of SDGFT, where inflation is driven by the dimensional flow from the UV fixed point to the IR attractor. We calculate the primordial spectral index $n_s \approx 0.967$ and the tensor-to-scalar ratio $r \approx 0.013$. We present the formal erratum correcting the legacy blue spectrum formula.

1 Introduction

Inflation is typically driven by an inflaton field. In SDGFT, it arises from the transition of the spectral dimension of spacetime.

2 Theoretical Formulation

Here we formulate the core mathematical relations governing the system:

$$n_s = 1 - \frac{2(2n-1)}{N_e(2n-1) + n} \approx 0.9671 \quad (1)$$

$$r = \frac{48(2n-1)^2}{[N_e(2n-1) + n]^2} \approx 0.0130 \quad (2)$$

3 CMB Observables

The scalar spectral index and tensor-to-scalar ratio are calculated using the Jordan-frame slow-roll equations for $f(R)$ gravity.

4 Conclusion

The predicted tensor-to-scalar ratio $r \approx 0.013$ will be tested by next-generation polarization experiments, providing a test of the theory.

References

- [1] D. A. Besemer, “Axiomatic Foundations of SDGFT,” SDGFT-2026-01 (in preparation).
- [2] D. A. Besemer, “Effective Spectral Dimension and the Banach Fixed-Point Theorem in SDGFT,” SDGFT-2026-04.
- [3] D. A. Besemer, `sdgft` Python package, <https://github.com/david-besemer/sdgft> (2026).